

E-Commerce Database Management System

1. Project Overview

The **E-Commerce Database Management System** is a SQL Server database project designed to manage the main operations of an e-commerce business.

The system manages customers, products, categories, suppliers, orders, order items, payments, shipments, coupons, and order status changes.

The project demonstrates database design, data management, business logic implementation, database optimization, and data visualization using Power BI.

2. Main Features

- Customer management
- Product and category management
- Supplier management
- Order and order item management
- Payment processing
- Shipment tracking
- Coupon and discount management
- Order status auditing
- Stock management
- Database functions
- Stored procedures with transactions and error handling
- Database triggers
- Indexing and query optimization
- Power BI dashboard for data visualization

3. Database Structure

The main database tables are:

- Categories
- Coupons
- Customers
- OrderItems
- Orders
- OrderStatusAudit
- Payments

- Products
- Shipments
- Suppliers

The database uses **Primary Keys, Foreign Keys, CHECK Constraints, UNIQUE Constraints, and DEFAULT Constraints** to maintain data integrity.

4. Database Setup

1. Open **SQL Server Management Studio (SSMS)**.
2. Execute the database and table creation script.
3. Execute the seed data script or import the provided CSV data.
4. Execute the business queries.
5. Execute the views, functions, stored procedures, and triggers.
6. Execute the indexes and optimization script.
7. Open the Power BI dashboard and connect it to the database.

5. Project Files

The project contains the following deliverables:

- [00_ERD_Diagram.pdf/png](#) – ERD and relational schema diagram.
- [01_Schema_DDL.sql](#) – Database and table creation scripts.
- [02_Seed_Data_DML.sql](#) – Sample data and data insertion scripts.
- [03_Business_Queries_DQL.sql](#) – Business and analytical queries.
- [04_Views_And_Programmability.sql](#) – Views, functions, stored procedures, and triggers.
- [05_Indexes_And_Optimization.sql](#) – Indexes, execution plans, and optimization.
- [Project_Report.pdf](#) – Complete project documentation.
- [README.md](#) – Project overview, setup instructions, and ERD overview.

6. ERD Overview

The Entity Relationship Diagram represents the relationships between the main entities in the e-commerce system.

Customers place orders, and each order contains multiple order items. Order items are associated with products, while products belong to categories and can be supplied by suppliers.

Orders are connected to payments and shipments, while coupons can be applied to orders. Order status changes are recorded in the OrderStatusAudit table.

The ERD provides a visual representation of the database structure and the relationships between its entities.

7. Technologies Used

- Microsoft SQL Server
- SQL Server Management Studio (SSMS)
- SQL
- Power BI
- CSV Data Import

8. Project Report

The [Project_Report.pdf](#) contains the project documentation, including business rules, assumptions, data import screenshots, execution plans, database optimization, and the Power BI dashboard.

